

Q. No.	Questions
	Unit 1
1.	In an electric vehicle battery cooling system, the temperature T satisfies $\frac{dT}{dt} = -k(T - T_a)$, where T_a is ambient temperature. Solve using separation of variables and evaluate the time required for the battery to reach a safe temperature given initial conditions.
2.	In a fluid drainage system, the height h of water in a tank satisfies $\frac{dh}{dt} = -k\sqrt{h}$ Apply separation of variables to compute the time taken for the tank to empty and interpret how system parameters affect drainage rate.
3.	In a wireless sensor system, signal strength S decays as $\frac{dh}{dt} = -k\sqrt{h}$ Solve the equation using separation of variables and analyze how initial signal strength influences transmission range.
4.	In a population model of bacteria growth in a bioreactor, $dP/dt = kP$ Solve using separation of variables and identify how doubling time depends on growth rate.
5.	A student solves $dx/dy = xy$ but separates variables incorrectly. Identify the correct separation step and compute the general solution.
6.	In a heat transfer model, $dT/dt = -kT$. Identify whether the equation is separable and justify.
7.	For the equation $dy/dx = x/y$, separate the variables and state the integrable form.
8.	In an electric vehicle, the battery pack experiences non-uniform heating during fast charging. The temperature rise is influenced by internal resistance and external cooling, leading to the model $\frac{dy}{dx} + 3y = 2x - 1$ where y represents temperature deviation and x represents time. Find the temperature distribution using Leibnitz method and analyze whether the system reaches thermal stability.

9.	In a robotic arm, position error is governed by feedback and actuator delay. The system is modeled as $x \frac{dy}{dx} + y(x+1) = 1$ where y is error and x is time. Convert, solve, and evaluate whether the system eliminates steady-state error.
10.	A damping system in machinery follows $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$ An engineer incorrectly assumes integrating factor e^{-2x}, leading to wrong results. Identify the mistake, compute correct integrating factor, and solve the equation.
11.	In a smart sensor, voltage output varies with environmental conditions and is modeled as $(x+1) \frac{dy}{dx} - y = e^{3x} (x+1)^2$ Transform into standard form and compute integrating factor.
12.	A communication system processes signals following $\frac{dy}{dx} + \frac{y}{x \log x} = \frac{2 \log x}{x \log x}$ Evaluate integrating factor and interpret the substitution used.
13.	In an RC circuit, voltage variation is modeled by $\frac{dy}{dx} + \frac{y}{x} = x^2$ Identify integrating factor and state its significance.
14.	In a chemical process, concentration change follows $\frac{dy}{dx} + 5y = 4x - 1$ Find integrating factor and interpret its role in simplifying solution.
15.	In an autonomous vehicle navigation system, the trajectory is represented by a family of curves $x^2 + y^2 = r^2$ where r varies depending on turning radius. Form the differential equation representing this motion and interpret its physical meaning.
16.	A vibrating system is described by displacement $y = a \sin(x + b)$ where a and b depend on system calibration. Form the differential equation by eliminating constants and evaluate its engineering significance.

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17.	A signal amplification system follows $y = ae^{4x} + be^{-x}$ where a, b depend on initial signal conditions. Form the differential equation and analyze system response behavior.
18.	A robot moves in straight lines described by $y = mx + c$ where slope varies with terrain. Form differential equation when both m and c are arbitrary and justify order.
19.	Fluid flow velocity is given by $y = \frac{a}{x} + b$ where constants depend on system pressure. Form the differential equation and interpret the result.
20.	A sensor output follows $y = e^x (a \cos x + b \sin x)$ Identify the order of resulting differential equation and justify.
21.	A sensor output follows $y = e^{bx} (a \cos x)$ Identify the order of resulting differential equation and justify.
22.	An EV battery initially at 120°C is placed in an environment at 30°C. After 10 minutes, its temperature drops to 80°C. Using Newton's law of cooling, formulate, solve, and estimate the temperature after 20 minutes. Also interpret system behavior.
23.	A heated metal rod cools from 200°C to 150°C in 5 minutes in a room at 25°C. Find the time required to reach 50°C and analyze cooling efficiency.
24.	A body of mass 5 lb is dropped from a height of 100 ft with zero velocity. Assuming no air resistance, find - an expression for the velocity of the body at any time t, - an expression for the position of the body at any time t,
25.	A sensor cools from 90°C to 70°C in 8 minutes in a 20°C room. Find the cooling constant k.
26.	A body is dropped from rest. Formulate the differential equation and find velocity.
27.	Apply Newton's law of cooling to formulate the differential equation for an object at 90°C placed in a surrounding of 25°C.
28.	Apply the concept of free fall to construct the differential equation and determine the velocity of a body dropped from rest.
29.	A radioactive sample initially has 80 mg. After 3 hours, it reduces to 60 mg. Formulate, solve, and evaluate the mass remaining after 6 hours.
30.	A bacterial culture grows from 500 to 1500 in 4 hours. Develop the model and predict population after 6 hours.
31.	A vehicle slows down in water where deceleration is proportional to velocity. Initial velocity is 20 m/s. After 5 seconds, velocity becomes 10 m/s. Formulate, solve, and interpret velocity after 10 seconds.

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32.	An investment of ₹10,000 grows to ₹15,000 in 5 years. Determine the growth constant and interpret doubling time.
33.	A drug concentration reduces to 75% in 2 hours. Analyze decay constant and estimate concentration after 5 hours.
34.	A motorboat slows down such that its retardation is proportional to velocity. Formulate the governing differential equation and identify its type.
35.	In a cooling system, interpret why temperature difference governs the rate of cooling.
36.	A viral infection spreads in a closed campus of 5000 students. Initially, 5 students are infected and after 4 days, 50 students are infected. Formulate the logistic model, determine constants, and evaluate infected population after 10 days.
37.	A fish population in a lake follows logistic growth with carrying capacity 2000. Initially, population is 100 and after 3 months it becomes 400. Develop the model and predict population after 8 months.
38.	A chemical reactor produces a substance limited by capacity 1000 units. Initially 50 units exist and after 2 hours it becomes 200 units. Formulate the logistic equation and analyze production after 6 hours.
39.	In a logistic system with carrying capacity 800, determine and interpret the population when growth rate is maximum.
40.	An engineer models growth as $dP/dt = kP^2$. Evaluate and correct the model using logistic principles.
41.	Explain why logistic growth is slower than exponential growth at later stages.
42.	Analyze why the logistic growth model initially behaves like exponential growth.
Unit 2	
1.	A spring-mass system is governed by $d^2y/dt^2 + 5dy/dt + 6y = 0$ Classify the equation, determine the complementary function, and interpret system behavior.
2.	An RLC circuit is modeled as $d^2i/dt^2 + 4di/dt + 4i = e^{(-t)}$ Classify the equation, determine the complementary function, and interpret system behavior.
3.	A control system follows $d^2y/dx^2 + 4dy/dx + 13y = 0$ Analyze roots, determine CF, and interpret system stability.
4.	Classify and justify the equation: $d^2y/dx^2 + 3y = \sin x$
5.	A system follows $d^2y/dx^2 + 4y = 0$, with $y(0) = 2$, $y'(0) = -4$. The complementary function is $y = C_1 \cos 2x + C_2 \sin 2x$. Determine the constants and interpret system behavior.
6.	Interpret the role of particular integral in second order systems in physical and engineering contexts.

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6.	Interpret the role of particular integral in second order systems in physical and engineering contexts.

7.	Explain the role of initial conditions in second order systems.
8.	A spring-mass system is governed by $d^2y/dt^2 + 5dy/dt + 6y = 0$ Formulate the auxiliary equation, determine the complementary function, and analyze system behavior.
9.	An RLC circuit satisfies $d^2i/dt^2 + 4di/dt + 4i = 0$ Determine complementary function and evaluate system response.
10.	A control system is modeled by $d^2y/dx^2 + 2dy/dx + 10y = 0$ Analyze roots, determine complementary function, and interpret oscillation.
11.	Analyze the equation $d^2y/dt^2 + 6dy/dt + 9y = 0$ and interpret the nature of motion.
12.	A vibration system is modeled by $d^2y/dt^2 + 9y = 0$ Identify the nature of roots and interpret system behavior.
13.	Interpret why real negative roots indicate stable systems.
14.	Identify the condition for overdamped system in second order models.
15.	An RLC circuit is modeled by $d^2i/dt^2 + 3di/dt + 2i = e^{-t}$ Formulate, determine CF and PI using undetermined coefficients, and construct the complete solution.
16.	A vibration system is governed by $d^2y/dt^2 + 4y = t^2$ Determine particular integral using undetermined coefficients and evaluate complete response.
17.	A control system is described by $d^2y/dx^2 + y = \cos 2x$ Determine the particular integral using another method.
18.	An electrical system is described by $d^2i/dt^2 + 3di/dt + 2i = e^{2t}$ Find the particular integral.
19.	A vibrating system is governed by $d^2y/dt^2 + 4y = \sec t$ Evaluate and compare the applicability of undetermined coefficients and variation of parameters.
20.	A vibration system is modeled by $d^2x/dt^2 + 3dx/dt + 2x = 5\cos t$ Identify and justify whether the system is homogeneous or non-homogeneous.
21.	In a forced mechanical system, interpret the significance of complete solution.

22.	A mass of 2 kg is attached to a spring with stiffness 50 N/m. Formulate the differential equation, determine angular frequency, and evaluate the motion equation if initial displacement is 0.3 m and initial velocity is 0.
23.	A spring system follows $x'' + 16x = 0$ Analyze amplitude, frequency, and evaluate total energy expression.
24.	An LC circuit is governed by $d^2q/dt^2 + 9q = 0$ Determine the solution and interpret periodic behavior.
25.	A mass-spring system satisfies $x'' + 25x = 0$ Analyze and interpret amplitude, frequency, and nature of motion.
26.	An LC circuit has equation $d^2q/dt^2 + (k/m)q = 0$ Evaluate the effect of increasing k on frequency.
27.	A spring-mass system in vibration analysis is modeled by a second-order differential equation is $\frac{d^2x}{dt^2} + \omega^2x = 0$. Interpret the type of motion.
28.	A mechanical oscillator completes 50 cycles in 10 seconds. Determine its frequency and interpret its engineering significance.
29.	A vehicle suspension system is modeled by $m \frac{d^2x}{dt^2} + c \frac{dx}{dt} + kx = 0$ with $m = 2$ kg, $c = 6$ Ns/m, $k = 5$ N/m. Formulate the differential equation, determine the nature of damping, and interpret system response.
30.	A vibration system satisfies $d^2x/dt^2 + 8dx/dt + 16x = 0$ Analyze the damping condition and evaluate system response.
31.	A system is governed by $d^2x/dt^2 + 10dx/dt + 21x = 0$ Determine the nature of roots and interpret system behavior.
32.	A system has equation $d^2x/dt^2 + 4dx/dt + 13x = 0$ Classify the system and interpret its motion.
33.	Interpret the role of damping co-efficient in damped vibration systems used in engineering structures.
34.	A system has equation $d^2x/dt^2 + 4dx/dt + 8x = 0$ Classify the system and interpret its motion.

35.	Interpret overdamped system behavior in engineering systems..
36.	A spring-mass-damper system is subjected to an external force $F(t) = 10\cos 2t$. Given $m = 1$ kg, $c = 4$ Ns/m, $k = 5$ N/m, formulate the differential equation, determine steady-state solution, and interpret system behavior.
37.	A structure vibrates under external force $F(t) = 8\cos \omega t$. Analyze and determine the condition for resonance and evaluate its effect.
38.	A forced system follows $x'' + 2x' + 10x = 5\cos 3t$ Determine amplitude response and analyze phase difference.
39.	Analyze and interpret the difference between transient and steady-state response in forced vibration systems.
40.	A spring-mass-damper system is governed by $x'' + 4x' + 5x = 10\cos 2t$ Determine the steady-state amplitude and interpret system behavior.
41.	A forced system satisfies $x'' + 6x' + 25x = 10\cos 4t$ Determine phase difference.
42.	A system has parameters $k = 16$, $c = 2$, $\omega = 3$, $F_0 = 6$. Compute steady-state amplitude.
Unit 3	
1.	An electric vehicle follows the trajectory $\vec{r}(t) = t^2\hat{i} + 3t\hat{j}$. Compute velocity and acceleration at $t = 2$. Analyze whether the motion is uniformly accelerated and justify.
2.	A robotic system path is given by $\vec{r}(t) = \cos t\hat{i} + \sin t\hat{j}$. Compute velocity and unit tangent vector at $t = \pi/2$. Interpret direction of motion.
3.	A drone path is $\vec{r}(t) = t^3\hat{i} + 2t^2\hat{j} + t\hat{k}$. Compute velocity, acceleration, and evaluate tangential component at $t = 1$.
4.	In a conveyor system, $\dot{\vec{r}}(t) = t^2\hat{i} + t^2\hat{j}$. Analyze whether direction of velocity changes with time.
5.	A student computes derivative of $\vec{r}(t) = t^2\hat{i} + e^t\hat{j}$ incorrectly. Identify error if $\vec{v} = 2t\hat{i} + te^t\hat{j}$ and correct it.
6.	A particle has $\dot{\vec{r}}(t) = 4t\hat{i} + 3t\hat{j}$. Compute speed and interpret motion.
7.	A system models only acceleration without velocity. Identify limitation.
8.	In an electric vehicle battery system, temperature distribution is modeled by $T(x, y, z) = x^2 + y^2 + z^2$. Compute the gradient and evaluate the direction of maximum temperature rise at point (1,1,1). Interpret its physical significance for thermal management and justify.



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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Approved Questions S2 Semester

22AI206/22AM206/22CS206/22IT206 – DIGITAL COMPUTER ELECTRONICS

Q. No.	Questions
	Unit 1
1.	A calculator device internally processes numbers in binary but must display outputs in a user-friendly format. The display has limited space, and binary representation appears lengthy. The system designer must choose between octal and hexadecimal formats for output representation while ensuring accuracy and minimal processing overhead. Determine the most appropriate number system for display and justify the decision based on efficiency and usability.
2.	A program converts binary values into decimal but fails when fractional values are present, producing incomplete results. The integer portion is processed correctly using positional weights, but the fractional portion is ignored. The developer must identify missing logic affecting correctness. Identify the error in the program and determine its effect on output accuracy.
3.	A telemetry system converts decimal data into binary and then into octal for transmission. Engineers observe incorrect outputs when fractional values are processed. The system uses standard conversion techniques but suffers from truncation and grouping inconsistencies, affecting final transmission accuracy. Diagnose the cause of incorrect outputs and infer how binary conversion errors impact octal representation.
4.	A processing unit converts inputs from multiple number systems into binary, performs operations, and converts results back into original formats. Improper sequencing has caused incorrect outputs in previous tests. The system must ensure proper data flow and accuracy. Formulate a correct sequence for number system conversion and infer the impact of incorrect sequencing.

5.	A system processes the decimal value 9.375 and converts it into binary and hexadecimal for computation and display. The conversion must follow positional representation rules without truncation. Determine binary and hexadecimal equivalents and justify each step logically.
6.	A computing system frequently converts large datasets between number systems. Binary ensures processing compatibility, while hexadecimal reduces storage size. Fractional conversions increase processing time. Assess trade-offs and recommend an efficient strategy for performance optimization.
7.	A communication system processes inputs in multiple number systems and converts them into binary for transmission. Outputs must be reconverted accurately with minimal overhead and high precision. Design a consistent model for number system conversion and justify its effectiveness.
8.	A digital calculator must represent negative numbers internally for performing arithmetic operations. The initial design used signed magnitude representation but faced inefficiencies in subtraction operations. The design team evaluates complement-based methods to improve performance and reduce hardware complexity. Determine the most appropriate representation method and justify the choice based on computational efficiency.
9.	A program implementing 1's complement subtraction produces incorrect results due to improper handling of carry. The developer notices missing logic when the addition generates an extra carry bit. Identify the fault and determine its impact on subtraction accuracy.

10.	A control system using 2's complement arithmetic produces incorrect outputs when adding large negative numbers. The system fails to detect overflow conditions, leading to incorrect interpretations of results. Diagnose the issue and infer how overflow affects system reliability.
11.	A processor performs subtraction using complement methods but occasionally produces incorrect results due to improper sequencing of operations. The system must ensure correct complement formation and addition. Formulate the correct sequence for complement-based subtraction and infer the impact of incorrect sequencing.
12.	A system computes $(-4) + (-2)$ using 4-bit 2's complement representation and must verify the correctness of the result within range limits. Determine the result using 2's complement and justify the steps logically.
13.	A processor design compares signed magnitude, 1's complement, and 2's complement for implementing arithmetic operations efficiently at scale. Assess trade-offs and recommend the best representation method.
14.	A digital system integrates signed number representation into arithmetic operations and must ensure consistency and accuracy across all computations. Design a consistent model for signed number representation and justify its effectiveness.
15.	A communication system must select one signal from multiple inputs based on control signals for transmission. The designer considers using either individual logic gates or a dedicated combinational module. The system requires minimal delay and efficient hardware usage. Determine the appropriate module for signal selection and justify the decision based on efficiency and performance.

